Penguins

2025-09-26

Introduction

This is a simple analysis of the Palmer Penguins dataset. This analysis will go over linear regression, data visualization, anova, and t-testing. If you would like at the code for this project, please visit the markdown file.

Libraries

Here are the libraries used for this viusualization and analysis.

```r
library(datasets)
library(ggplot2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
library(tibble)
library(palmerpenguins)
library(BSDA)
```

```
## Loading required package: lattice
##
## Attaching package: 'BSDA'
##
## The following object is masked from 'package:datasets':
##
##     Orange
```

```r
library(ggcorrplot)
```

```
## Warning: package 'ggcorrplot' was built under R version 4.4.3
```

Our Dataset

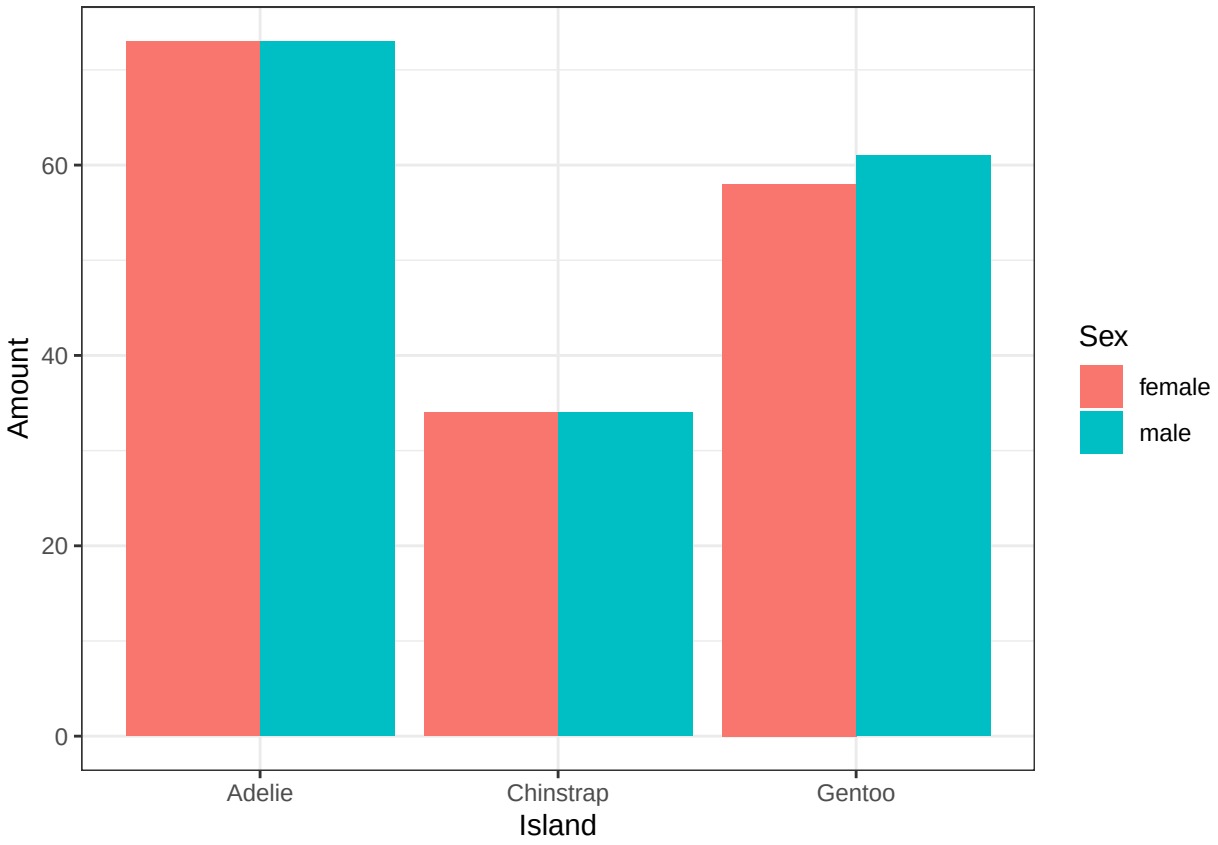
This dataset contains 344 observations with 7 factors:

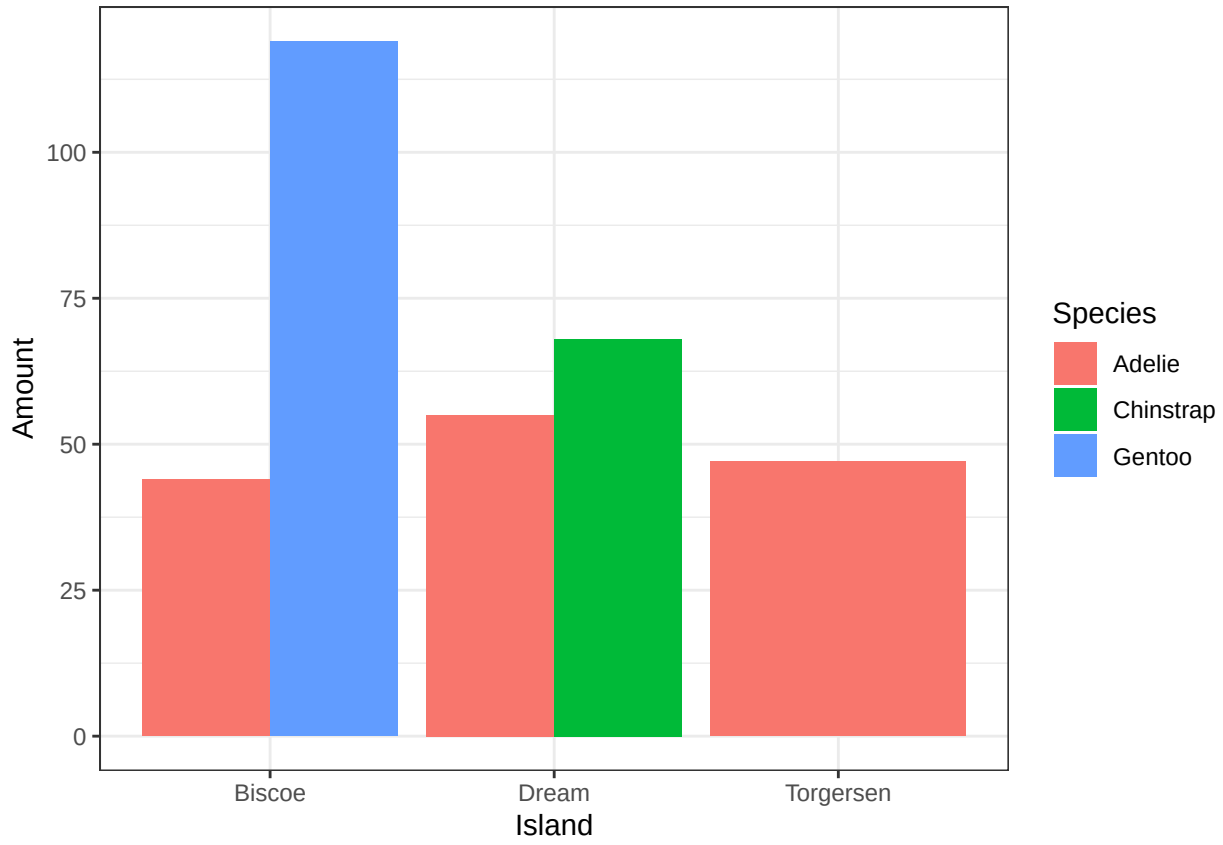
Species: Denotes penguin species
Island: Denotes an island in the Palmer Archipelago where the penguin was observed.
Bill_length_mm: Bill length in millimeters.
Bill_depth_mm: Bill Depth in millimeters.
Flipper_length_mm: Flipper Length in millimeters.
Body_mass_g: Body mass in grams.
Sex: Denotes penguin sex (Male/Female).
Year: Denotes study year(2007,2008, or 2009)

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

Initial Visualizations

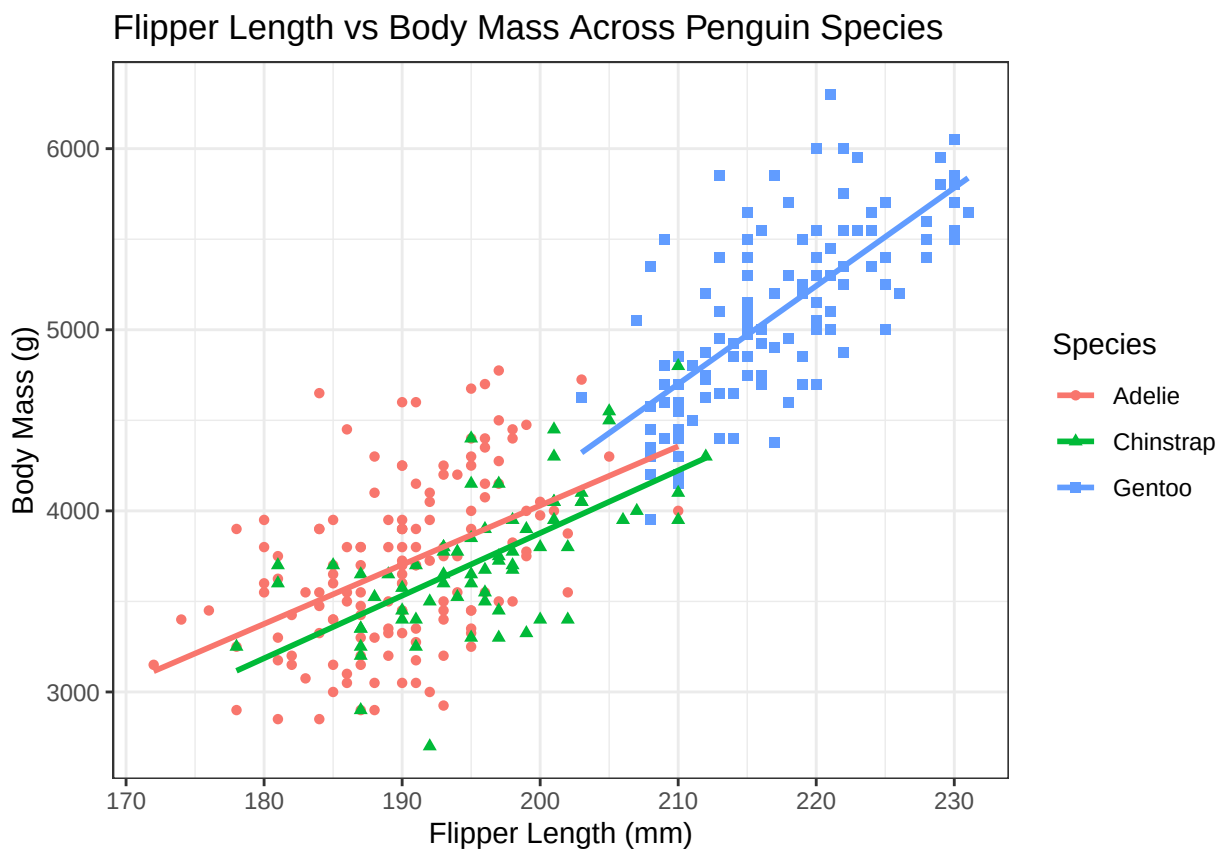
Before we dive into some regression analysis, let's get an overall understanding of what our data looks like and how it is distributed. With these graphs, we can see that we have a roughly equal amount of male and female penguins. Adelie penguins are the most plentiful, and are also the only species found on every island.


Amount
60
40
20
0
Adelie
Chinstrap
Gentoo
Island
Sex
female
male

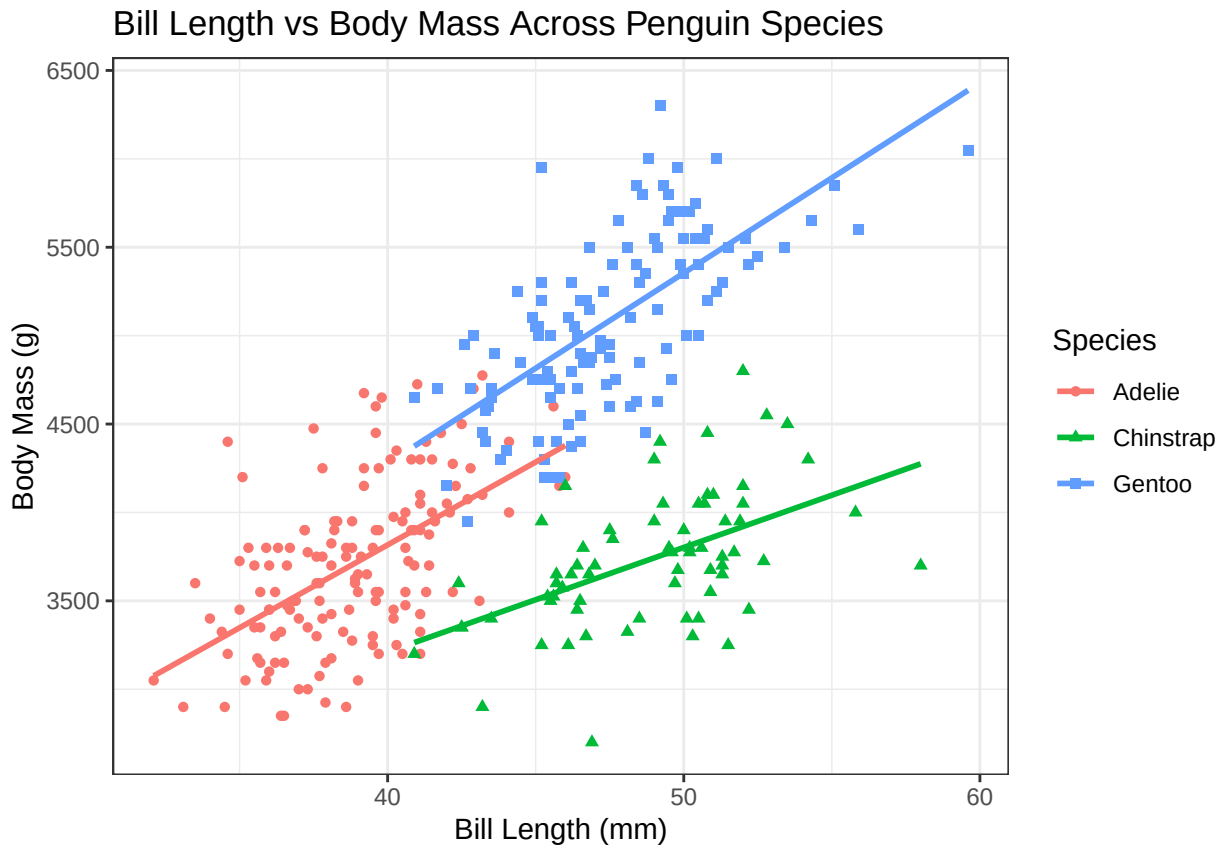


Suppose we want to analyze if there analyze the factors behind penguin mass. The following are graphs depicting the relationship between body dimensions and body mass across all species of penguins. Adelie and Chinstrap Penguins seem to have similar dimensions in regards to flipper length and bill depth. We can also see that there seems to be some positive relationship between body dimenstions and body mass.

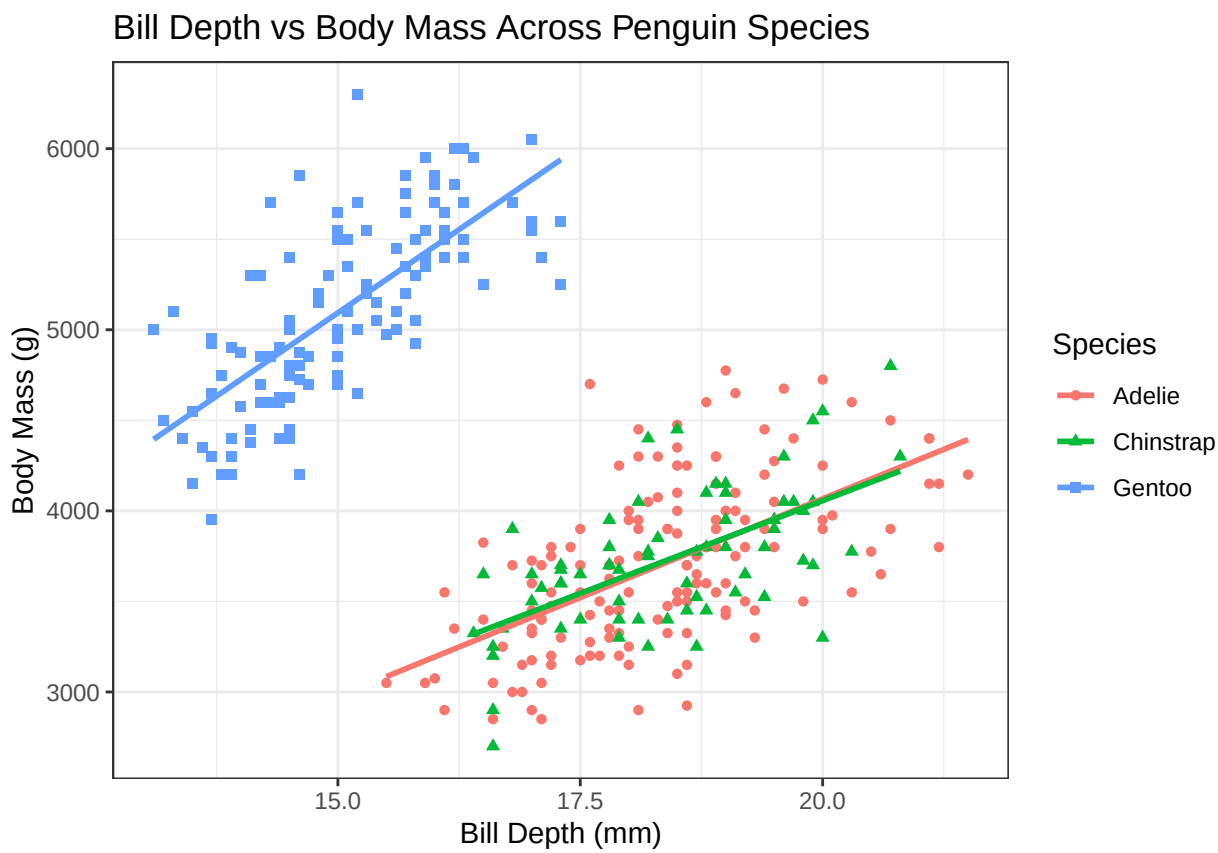
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```



Model Creation and Analysis

Now let's create some models that can help us predict certain dimensions or features of a given penguin. For these models, let us assume a significance level of 0.05. Suppose we wanted to predict a penguin's body mass using its other body dimensions. Since different species will have different attributes, we should focus on one species at a time. In the above visualizations, we can observe that the Gentoo are isolated from the other species regarding body mass, let's focus on them for now. Let's start with a simple regression analysis using flipper length. In this reduced model, we can see that flipper length is a significant predictor for body mass, with an extremely small p-value well below our significance level of 0.05. We can reject the null and assume the alternative is true: flipper length is a significant predictor for body mass. Our model is as follows:

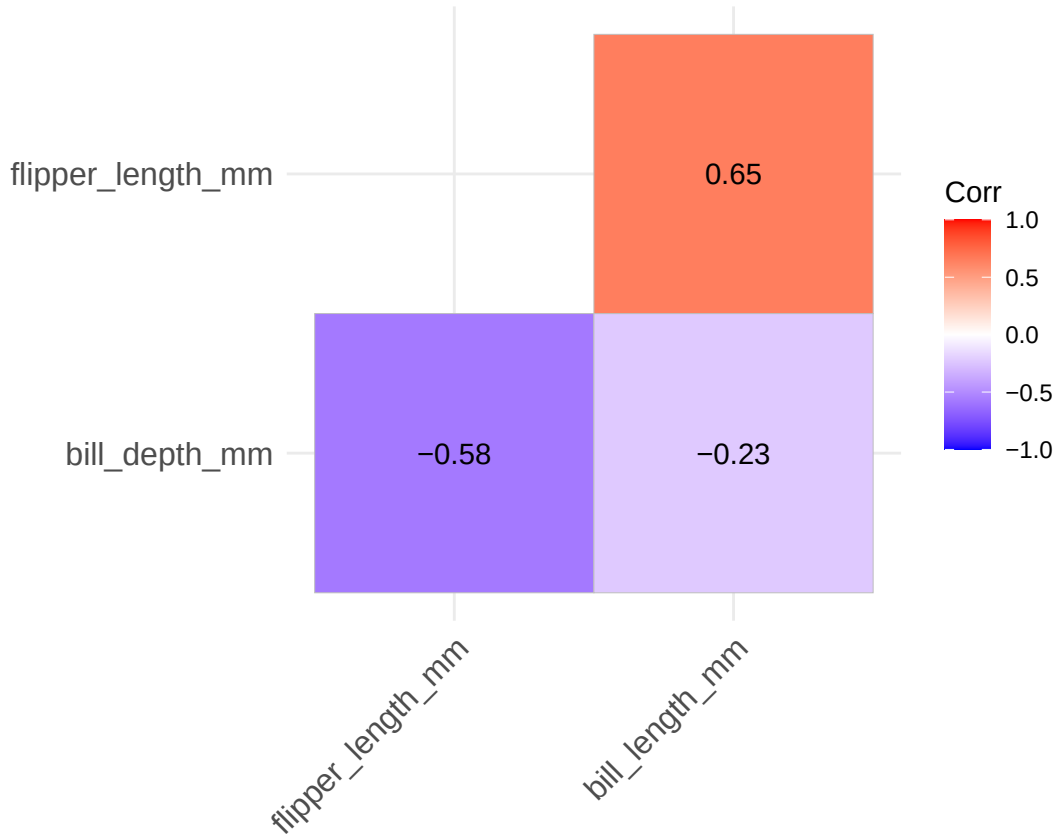
$$BodyMass = -6674.20 + 54.12 FlipperLength$$

With every 1mm increase with flipper length for a Gentoo penguin, we can expect body mass to increase by 54 grams.

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = gentoo.penguins)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -704.69 -244.29  -58.87  161.98 1003.65
##
```

```
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -6674.204   1075.436  -6.206 8.51e-09 ***
## flipper_length_mm     54.165      4.948  10.946  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 354 on 117 degrees of freedom
## Multiple R-squared:  0.506,  Adjusted R-squared:  0.5017
## F-statistic: 119.8 on 1 and 117 DF,  p-value: < 2.2e-16
```

Suppose we wanted to include more factors into this model and make it a multilinear model, such as Bill Length and Bill Depth. In order to do this, we have to establish that these variables have a linear relationship with body mass, and share low multicolinearity. Our scatterplots above demonstrate the latter, so we should verify the former through a correlation matrix

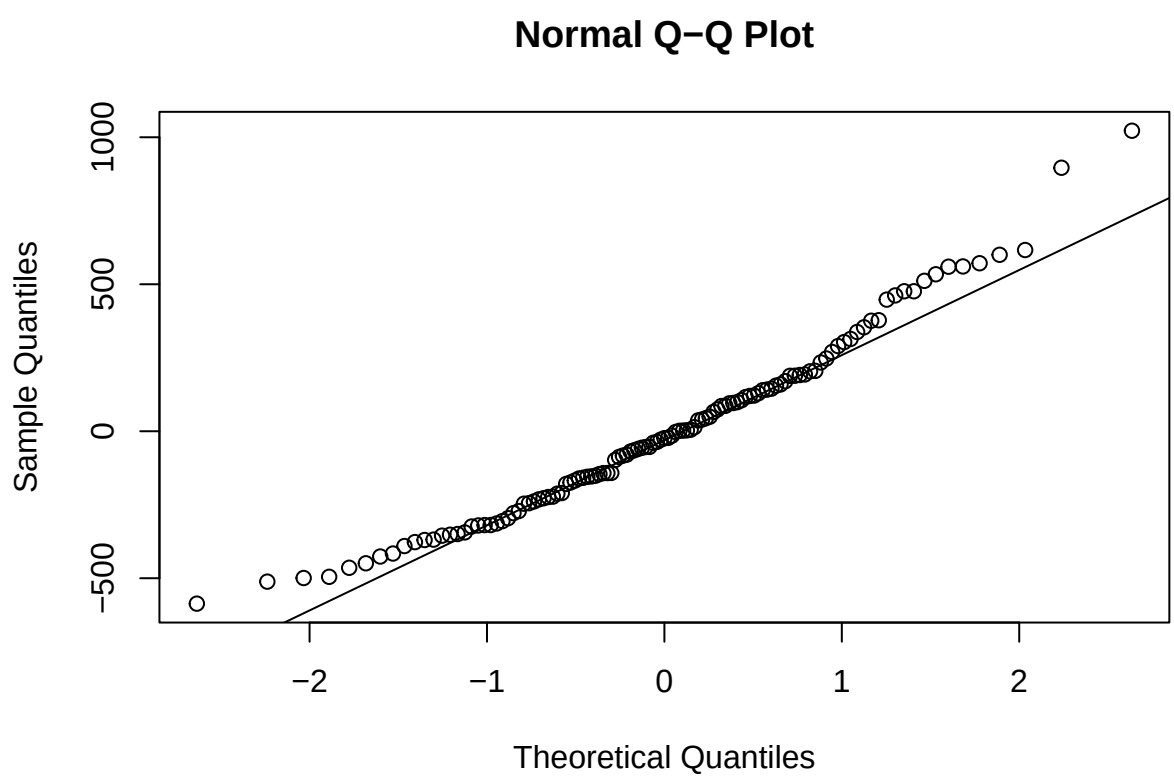


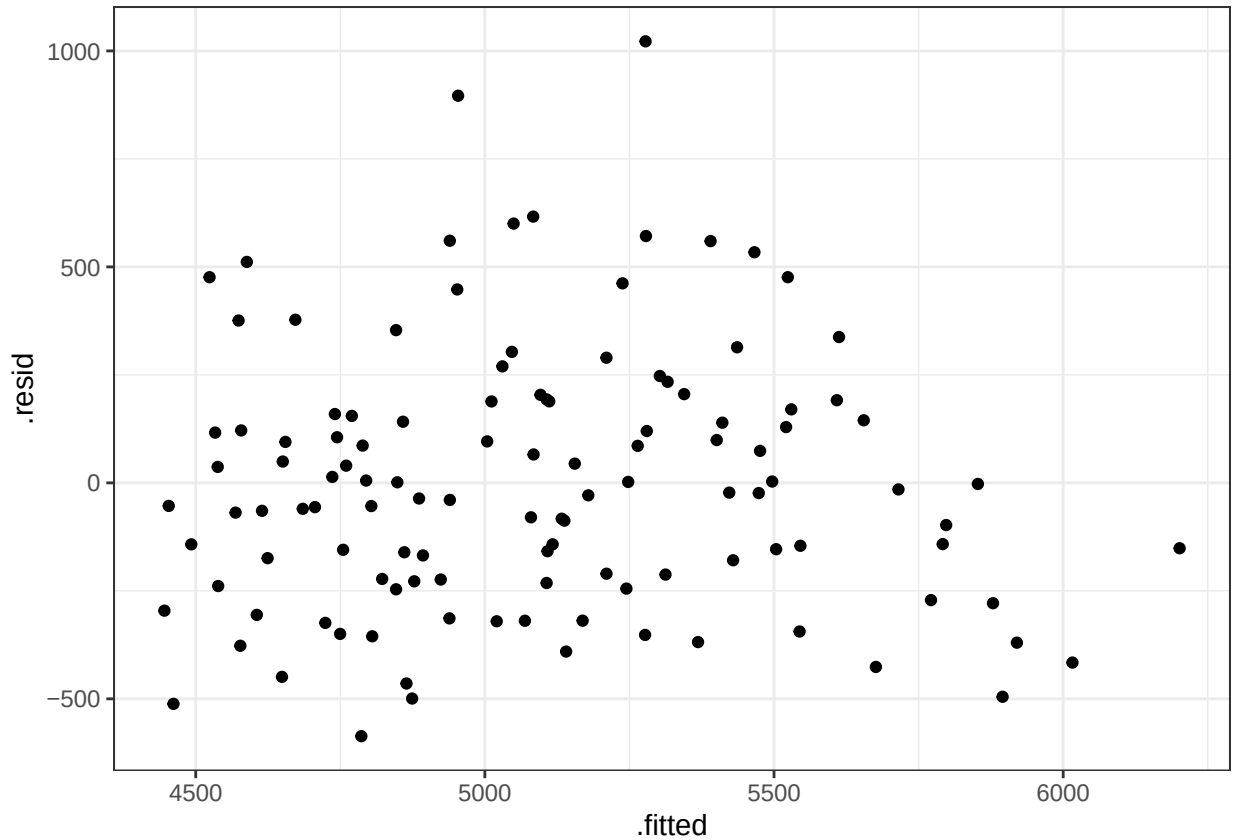
Bill Length and Flipper Length have a relatively high correlation of 0.65. This is not so high that we exclude Bill Length in a full model, but it is possible that it could affect the results in some way and we should keep it in mind. We also have to test for normality and homoscedasticity in our full model. We can do this by using a qq plot and plotting our fitted values against our residual ones.

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + bill_depth_mm +
##     bill_length_mm, data = gentoo.penguins)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -586.42 -225.83  -23.64  164.71 1022.17
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -4444.415   1009.081  -4.404 2.39e-05 ***
## flipper_length_mm    23.259      6.639   3.503 0.000655 ***
## bill_depth_mm       180.005     43.818   4.108 7.51e-05 ***
## bill_length_mm       37.517     13.091   2.866 0.004947 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 309.8 on 115 degrees of freedom
## Multiple R-squared:  0.6281, Adjusted R-squared:  0.6184
## F-statistic: 64.73 on 3 and 115 DF,  p-value: < 2.2e-16
```

```
## Analysis of Variance Table
##
## Model 1: body_mass_g ~ flipper_length_mm
## Model 2: body_mass_g ~ flipper_length_mm + bill_depth_mm + bill_length_mm
##   Res.Df      RSS Df Sum of Sq     F    Pr(>F)
## 1    117 14660505
## 2    115 11036708  2   3623798 18.88 8.123e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```





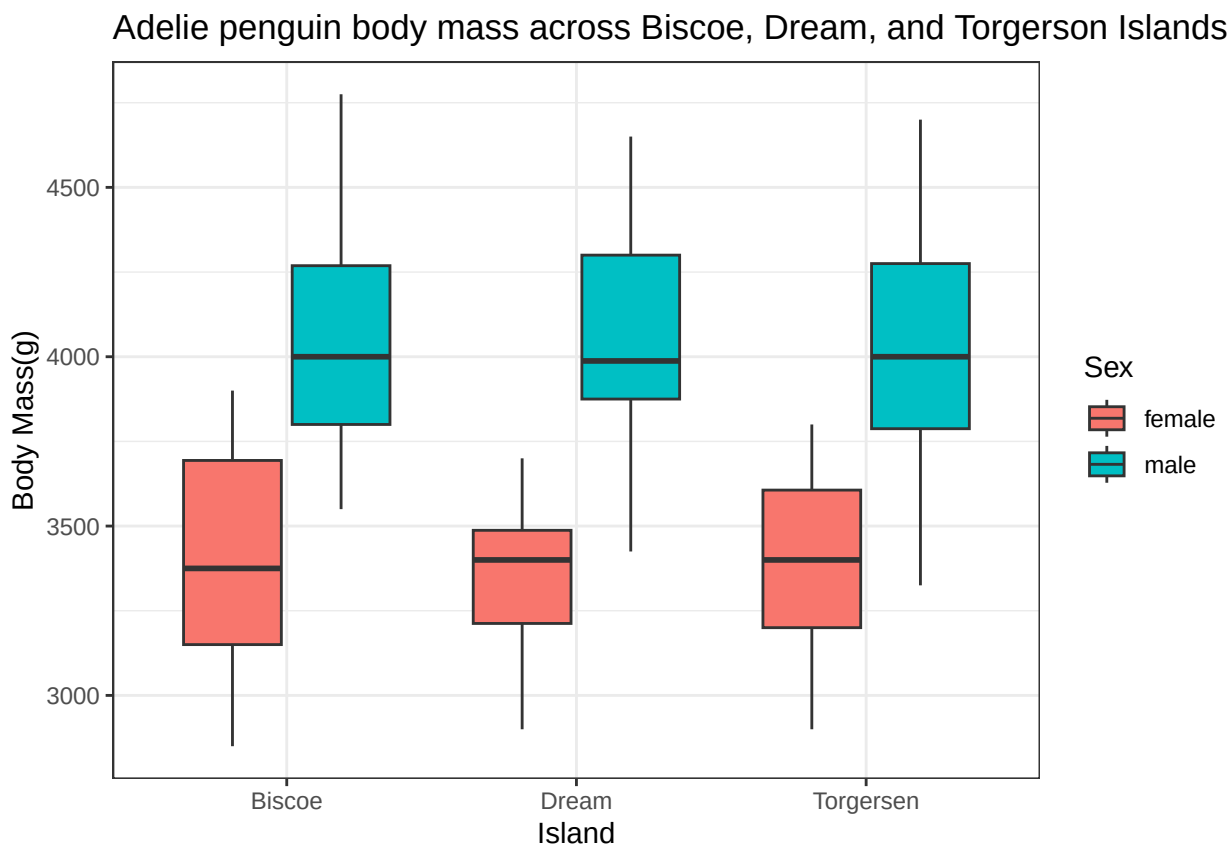
These results are promising. According to our anova testing, we result in a p-value well below our significance level of 0.05, meaning we can reject the null and assume that the alternative is true: Our full model is a better fit than our reduced one. Our full model also has significant p-values for all of our variables. We are also able to establish multivariate normality and homoscedasticity. Our qq plots lie across a straight line indicating that our residuals are normally distributed, and our fitted vs residual plot has little to no discernible pattern, indicitating that these residuals have constant variance along every point in our model. This is all enough suggestion to abandon our previous model and use our full model as a predictor for body mass. The resulting model is as follows:

$$BodyMass = -4444.42 + 23.26FlipperLength + 180.01BillDepth + 37.52BillLength$$

This means that for every 1mm increase in Flipper Length, Bill Depth, and Bill Length, we can expect approximately a 23, 180, and 38 gram increase in body mass, respectively, when holding all other variables constant. If we have a penguin with a flipper length of 220mm, bill depth of 13 mm, and bill length of 51mm, we can expect a body mass of -4444.42 + 23.26(220) + 180.01(13) + 37.52(51) = 4926.3 grams.

```
## # A tibble: 6 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>           <dbl>         <dbl>             <int>       <int>
## 1 Gentoo  Biscoe           46.1          13.2               211        4500
## 2 Gentoo  Biscoe           50            16.3               230        5700
## 3 Gentoo  Biscoe           48.7          14.1               210        4450
## 4 Gentoo  Biscoe           50            15.2               218        5700
## 5 Gentoo  Biscoe           47.6          14.5               215        5400
## 6 Gentoo  Biscoe           46.5          13.5               210        4550
## # i 2 more variables: sex <fct>, year <int>
```

Suppose we wanted to compare the mass of the same species penguins across islands and sex. Perhaps the habitat of each island could have an effect on the penguins, causing them to be larger, or male penguins are on average larger than female penguins, for example. For this analysis, let's focus on only Adelie penguins,since they are the only species found across all three islands. From this boxplot,we can see that there initially seems to be quite a different in mass between male and female penguins, and only a small difference between islands.



Using ANOVA in R for body mass between islands confirms these results: we result in a p-value well above our significance level of 0.05. We fail to reject the null and cannot confirm the alternative to be true. There is not enough evidence to assume that the island has an effect on the Adelie penguin's mass.

```
##              Df   Sum Sq Mean Sq F value Pr(>F)
## island        2     2064    1032   0.005  0.995
## Residuals   143 30496138  213260
```

Now let's see if there is a significant difference in mass for the sex of an Adelie penguin. We can use a simple t-test for this. In these results, we can see that the mean mass in grams for males is significantly larger than females. We additionally have a p-value of nearly 0, meaning that we reject the null and confirm the alternative as true. There is a statistically significant difference between the mass of male and female adelie penuins.

```
##
##  Welch Two Sample t-test
##
## data:  body_mass_g by sex
## t = -13.126, df = 135.69, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group female and group male is not equal to
## 95 percent confidence interval:
```

```
##  -776.3012 -573.0139
## sample estimates:
## mean in group female   mean in group male
##             3368.836             4043.493
```

Conclusion

This was an exploratory analysis of the palmer penguins dataset where we built a linear model that could predict a penguin's body mass given its body dimensions, as well as analyzed the difference between penguin populations between islands and penguin sex. Further research of this dataset could include using logistic regression or creating a ML classifier to identify a penguin species. One thing that should ne noted about this dataset is that it is quite small. The original dataset has only a few hundred observations. The methods used in this project and especially the creation of an ML classifier would result in more accurate models if used on a larger sample size that could more accurately represent the total penguin population.